

Luiss

Libera Università Internazionale
degli Studi Sociali Guido Carli

Homework Exercises on Binary Numbers and Boolean Logic

Introduction to Computer Programming

Management and Artificial Intelligence

LUISS



Binary Number Exercises

Decimal 387 to Binary

Convert the decimal number **387** to binary:

$$387 \div 2 = 193 \text{ (rem 1)}$$

$$193 \div 2 = 96 \text{ (rem 1)}$$

$$96 \div 2 = 48 \text{ (rem 0)}$$

$$48 \div 2 = 24 \text{ (rem 0)}$$

$$24 \div 2 = 12 \text{ (rem 0)}$$

$$12 \div 2 = 6 \text{ (rem 0)}$$

$$6 \div 2 = 3 \text{ (rem 0)}$$

$$3 \div 2 = 1 \text{ (rem 1)}$$

$$1 \div 2 = 0 \text{ (rem 1)}$$

Result: $(110000011)_2$

Binary 110110101011 to Decimal

Convert the binary number **110110101011** to decimal:

$$\begin{aligned}(110110101011)_2 &= 1 \cdot 2^{11} + 1 \cdot 2^{10} + 0 \cdot 2^9 + 1 \cdot 2^8 + 1 \cdot 2^7 + 0 \cdot 2^6 + 1 \cdot 2^5 + 0 \cdot 2^4 + 1 \cdot 2^3 \\ &= 2048 + 1024 + 0 + 256 + 128 + 0 + 32 + 0 + 8 + 0 + 2 + 1 = (3499)_{10}\end{aligned}$$

Decimal 2048 to Hexadecimal

Convert the decimal number **2048** to hexadecimal:

$$2048 \div 16 = 128 \text{ remainder } 0$$

$$128 \div 16 = 8 \text{ remainder } 0$$

$$8 \div 16 = 0 \text{ remainder } 8$$

Reading the remainders from bottom to top: 800

Therefore, $2048_{10} = 800_{16}$

Hex BADCODE to Binary

Convert the hexadecimal number **BADCODE** to binary:

Convert each hexadecimal digit to its 4-bit binary equivalent:

- $B_{16} = 1011_2$
- $A_{16} = 1010_2$
- $D_{16} = 1101_2$
- $C_{16} = 1100_2$
- $0_{16} = 0000_2$
- $D_{16} = 1101_2$
- $E_{16} = 1110_2$

Therefore: $BADC0DE_{16} = 1011101011011100000011011110_2$

Text TOTTI to Binary

Convert the text "TOTTI" to binary using the ASCII table

- T = 84 in ASCII = 01010100 in binary
- O = 79 in ASCII = 01001111 in binary
- T = 84 in ASCII = 01010100 in binary
- T = 84 in ASCII = 01010100 in binary
- I = 73 in ASCII = 01001001 in binary

Binary to text

Convert 0100110001010101010010010101001101010011 into text using ASCII:

Breaking the binary string into 8-bit chunks:

- 01001100 -> 76 in decimal -> 'L' in ASCII
- 01010101 -> 85 in decimal -> 'U' in ASCII
- 01001001 -> 73 in decimal -> 'I' in ASCII
- 01010011 -> 83 in decimal -> 'S' in ASCII
- 01010011 -> 83 in decimal -> 'S' in ASCII

So the text is: LUISS

Boolean Logic Exercises

Formula $(A \wedge \neg B \wedge C) \vee (\neg A \wedge B \wedge \neg C) \vee (A \wedge B \wedge C)$

A	B	C	!A	!B	!C	A$\wedge$!B$\wedge$C	!A$\wedge$B$\wedge$!C	A$\wedge$B$\wedge$C	(A$\wedge$!B$\wedge$C)$\vee$(!A$\wedge$B$\wedge$!C)$\vee$(A$\wedge$B$\wedge$C)
0	0	0	1	1	1	0	0	0	0
0	0	1	1	1	0	0	0	0	0
0	1	0	1	0	1	0	1	0	1
0	1	1	1	0	0	0	0	0	0
1	0	0	0	1	1	0	0	0	0
1	0	1	0	1	0	1	0	0	1
1	1	0	0	0	1	0	0	0	0
1	1	1	0	0	0	0	0	1	1

Formula $(\neg A \wedge \neg B \wedge \neg C) \vee (A \wedge B \wedge C) \vee (A \wedge \neg B \wedge \neg C)$

A	B	C	!A	!B	!C	!A$\wedge$!B$\wedge$!C	A$\wedge$B$\wedge$C	A$\wedge$!B$\wedge$!C	(!A$\wedge$!B$\wedge$!C)$\vee$(A$\wedge$B$\wedge$C)$\vee$(A$\wedge$!B$\wedge$!C
0	0	0	1	1	1	1	0	0	1
0	0	1	1	1	0	0	0	0	0
0	1	0	1	0	1	0	0	0	0
0	1	1	1	0	0	0	0	0	0
1	0	0	0	1	1	0	0	1	1
1	0	1	0	1	0	0	0	0	0
1	1	0	0	0	1	0	0	0	0
1	1	1	0	0	0	0	1	0	1

Formula $(A \wedge B) \vee (\neg A \wedge \neg B \wedge C) \vee (A \wedge \neg B \wedge \neg C)$

A	B	C	!A	!B	!C	A∧B	!A∧!B∧C	A∧!B∧!C	(A∧B)∨(!A∧!B∧C)∨(A∧!B∧!C)
0	0	0	1	1	1	0	0	0	0
0	0	1	1	1	0	0	1	0	1
0	1	0	1	0	1	0	0	0	0
0	1	1	1	0	0	0	0	0	0
1	0	0	0	1	1	0	0	1	1
1	0	1	0	1	0	0	0	0	0
1	1	0	0	0	1	1	0	0	1
1	1	1	0	0	0	1	0	0	1

Formula $(A \wedge B \wedge C) \vee (\neg A \wedge \neg B \wedge \neg C)$ is equivalent to what?

A	B	C	$A \wedge B \wedge C$	$\neg A$	$\neg B$	$\neg C$	$\neg A \wedge \neg B \wedge \neg C$	$(A \wedge B \wedge C) \vee (\neg A \wedge \neg B \wedge \neg C)$
0	0	0	0	1	1	1	1	1
0	0	1	0	1	1	0	0	0
0	1	0	0	1	0	1	0	0
0	1	1	0	1	0	0	0	0
1	0	0	0	0	1	1	0	0
1	0	1	0	0	1	0	0	0
1	1	0	0	0	0	1	0	0
1	1	1	1	0	0	0	0	1

Formula $(A \vee B) \wedge \neg(A \wedge B)$ is a tautology?

A	B	$A \vee B$	$A \wedge B$	$\neg(A \wedge B)$	$(A \vee B) \wedge \neg(A \wedge B)$
0	0	0	0	1	0
0	1	1	0	1	1
1	0	1	0	1	1
1	1	1	1	0	0

No, it is not a tautology! This is actually the XOR operation.

Formula $\neg A \wedge \neg B \wedge A$ is a contradiction?

A	B	!A	!B	!A∧!B∧A
0	0	1	1	0
0	1	1	0	0
1	0	0	1	0
1	1	0	0	0

Yes, it is a contradiction! It requires A to be both true and false simultaneously.

